Task 7: Develop Pig scripts for data processing.

Date of the Session: ----/-----/------ **Time of the Session: 10:20A.M. to 1:00P.M.**

Pre Lab:

1. What is Apache Pig and what role does it play in Big Data processing?

2. Differentiate between Pig Latin and SQL.

3. What are the various execution modes available in Pig?

4. Elucidate the fundamental Pig Latin operators utilized for data processing.

In-Lab:

1. Execute Pig Scripts for Data Processing.

Initially we are going to create some directories to store data in PIG using Hadoop. The details and marks directories are created using mkdir command, now we have created some text files to work on the data. Then we are going to perform sort, join on the data in PIG.

GROUPING:

Create a **student.txt** file using gedit and enter student name, age, and GPA data.

```
[cloudera@quickstart ~]$ gedit student.txt
[cloudera@quickstart ~]$ cat student.txt
rose,18,4.0f
jasmine,19,3.8f
sakura,20,3.9f
lily,18,3.8f
```

Create a directory in HDFS

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /tmp/student
```

Upload the student.txt file into HDFS

```
[cloudera@quickstart ~]$ hdfs dfs -put student.txt /tmp/student
```

Start the **PIG interactive shell** using the command pig

>pig

```
[cloudera@quickstart ~]$ pig
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more info.
2026-03-09 22:29:09,439 [main] INFO  org.apache.pig.Main - Apache Pig version 0.12.0-cdh5.13.0 (rexported) compiled O
ct 04 2017, 11:09:03
2026-03-09 22:29:09,439 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/cloudera/pig_177312054941
6.log
2026-03-09 22:29:09,460 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/cloudera/.pigbootup n
ot found
2026-03-09 22:29:09,944 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:29:09,944 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:29:09,944 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to h
adoop file system at: hdfs://quickstart.cloudera:8020
2026-03-09 22:29:11,069 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:29:11,069 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to m
ap-reduce job tracker at: localhost:8021
2026-03-09 22:29:11,069 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:29:11,133 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:29:11,134 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:29:11,216 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:29:11,217 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:29:11,290 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
```

Load the student dataset using:

grunt> A = LOAD '/tmp/student/student.txt/' USING PigStorage (',') AS (name:chararray,age:int,gpa:float);

```
grunt> A = LOAD '/tmp/student/student.txt'
>> USING PigStorage(',')
>> AS (name:chararray, age:int, gpa:float);
grunt> [cloudera@quickstart ~]$
```

Display the loaded data

grunt> DUMP A

```
grunt> DUMP A;
2026-03-09 22:41:28,488 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNK
NOWN
2026-03-09 22:41:28,530 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[
AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, Limi
tOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, Pus
hUpFilter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOpti
mizer]}
2026-03-09 22:41:28,635 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File c
oncatenation threshold: 100 optimistic? false
2026-03-09 22:41:28,650 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer
 - MR plan size before optimization: 1
2026-03-09 22:41:28,650 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer
 - MR plan size after optimization: 1
2026-03-09 22:41:28,765 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.
0:8032
2026-03-09 22:41:28,922 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the
 job
2026-03-09 22:41:28,970 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.b
uffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-03-09 22:41:28,970 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler
- mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-03-09 22:41:28,970 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is dep
```

```
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTIme      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime
AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1771910702594_0002  1       0       3       3       3       3       n/a     n/a     n/a     n/a     A       MAP_O
NLY     hdfs://quickstart.cloudera:8020/tmp/temp1519453116/tmp2017000289,

Input(s):
Successfully read 4 records (431 bytes) from: "/tmp/student/student.txt"

Output(s):
Successfully stored 4 records (75 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp1519453116/tmp2017000289"

Counters:
Total records written : 4
Total bytes written : 75
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1771910702594_0002


2026-03-09 22:41:49,192 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher -
 Success!
2026-03-09 22:41:49,193 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:41:49,193 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:41:49,193 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... wi
ll not generate code.
2026-03-09 22:41:49,203 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to pro
cess : 1
2026-03-09 22:41:49,203 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input path
s to process : 1
(rose,18,4.0)
(jasmine,19,3.8)
(sakura,20,3.9)
(lily,18,3.8)
```

Group the student data based on age

grunt>B = GROUP A BY age;

```
grunt> B = GROUP A BY age;
2026-03-09 22:42:12,818 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:42:12,818 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
```

grunt> DUMP B

```
grunt> DUMP B
2026-03-09 22:42:34,383 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GRO
UP_BY
2026-03-09 22:42:34,384 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[
AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, Limi
tOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, Pus
hUpFilter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOpti
mizer]}
2026-03-09 22:42:34,389 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File c
oncatenation threshold: 100 optimistic? false
2026-03-09 22:42:34,394 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer
 - MR plan size before optimization: 1
2026-03-09 22:42:34,394 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer
 - MR plan size after optimization: 1
2026-03-09 22:42:34,405 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.
0:8032
2026-03-09 22:42:34,406 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the
 job
2026-03-09 22:42:34,424 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler
- mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-03-09 22:42:34,424 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler
- Reduce phase detected, estimating # of required reducers.
2026-03-09 22:42:34,424 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler
- Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-03-09 22:42:34,426 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEst
imator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=57
2026-03-09 22:42:34,426 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler
```

```
Counters:
Total records written : 3
Total bytes written : 90
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1771910702594_0003


2026-03-09 22:42:58,087 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher -
 Success!
2026-03-09 22:42:58,087 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 22:42:58,087 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 22:42:58,089 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... wi
ll not generate code.
2026-03-09 22:42:58,093 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to pro
cess : 1
2026-03-09 22:42:58,093 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input path
s to process : 1
(18,{(lily,18,3.8),(rose,18,4.0)})
(19,{(jasmine,19,3.8)})
(20,{(sakura,20,3.9)})
```

JOIN:

gedit data1.txt

```
grunt> quit
[cloudera@quickstart ~]$ gedit data1.txt
[cloudera@quickstart ~]$ cat data1.txt
1,2,3
4,5,6
```

gedit data2.txt

```
[cloudera@quickstart ~]$ gedit data2.txt
[cloudera@quickstart ~]$ cat data2.txt
1,3
4,6
```

hdfs dfs -mkdir csec

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir csec
```

hdfs dfs -put data1.txt csec

```
[cloudera@quickstart ~]$ hdfs dfs -put data1.txt csec
```

hdfs dfs -put data2.txt csec

```
[cloudera@quickstart ~]$ hdfs dfs -put data2.txt csec
```

```
[cloudera@quickstart ~]$ hdfs dfs -ls csec
Found 2 items
-rw-r--r--   1 cloudera cloudera         12 2026-03-09 22:57 csec/data1.txt
-rw-r--r--   1 cloudera cloudera          8 2026-03-09 22:59 csec/data2.txt
```

gedit join.pig

```
[cloudera@localhost ~]$ gedit join.pig
```

pig

```
[cloudera@quickstart ~]$ pig
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more info.
2026-03-09 23:04:35,165 [main] INFO  org.apache.pig.Main - Apache Pig version 0.12.0-cdh5.13.0
ct 04 2017, 11:09:03
2026-03-09 23:04:35,165 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/clo
```

grunt> D = LOAD 'csec/data1.txt' USING PigStorage(',') AS (a1:int, a2:int, a3:int);

```
grunt> D = LOAD 'csec/data1.txt' USING PigStorage(',') AS
>> (a1:int,a2:int,a3:int);
```

grunt> DUMP D

```
grunt> DUMP D
2026-03-09 23:06:33,908 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features u
NOWN
2026-03-09 23:06:33,942 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimi
```

```
ll not generate code.
2026-03-09 23:06:53,893 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - To
cess : 1
2026-03-09 23:06:53,893 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedU
s to process : 1
(1,2,3)
(4,5,6)
```

grunt> DD = LOAD 'csec/data2.txt' USING PigStorage(',') AS (b1:int, b2:int);

```
grunt> DD = LOAD 'csec/data2.txt' USING PigStorage(',') AS (b1:int,b2:int);
2026-03-09 23:08:15,702 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 23:08:15,702 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
```

grunt> DUMP DD

```
grunt> DUMP DD
```

```
ll not generate code.
2026-03-09 23:09:14,237 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - To
cess : 1
2026-03-09 23:09:14,237 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedU
s to process : 1
(1,3)
(4,6)
```

Perform a **JOIN operation** on both datasets

```
grunt> X = JOIN D BY a1,DD BY b1;
```

grunt> DUMP X

```
grunt> DUMP X
```

```
2026-03-09 23:10:38,855 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematu
ll not generate code.
2026-03-09 23:10:38,859 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - To
cess : 1
2026-03-09 23:10:38,859 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedU
s to process : 1
(1,2,3,1,3)
(4,5,6,4,6)
```

Perform **UNION operation** on both datasets

grunt> UNION D,DD;

```
grunt> Y= UNION D,DD;
grunt> DUMP Y
2026-03-09 23:14:12,496 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features u
ON
2026-03-09 23:14:12,496 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimi
AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, Impli
tOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushD
```

grunt> DUMP

```
2026-03-09 23:14:36,279 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - To
cess : 2
2026-03-09 23:14:36,279 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedU
s to process : 2
(1,2,3)
(4,5,6)
(1,3)
(4,6)
```

Write a basic PIG script that performs a series of operations on the loaded data.

```
[cloudera@quickstart ~]$ gedit join.pig
[cloudera@quickstart ~]$ cat join.pig
D = LOAD 'csec/data1.txt' USING PigStorage(',') AS (a1:int,a2:int,a3:int);
DD = LOAD 'csec/data2.txt' USING PigStorage(',') AS (b1:int,b2:int);
X = JOIN D BY a1, DD BY b1;
DUMP X;
```

```
[cloudera@quickstart ~]$ pig join.pig
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more info.
2026-03-09 23:15:56,230 [main] INFO  org.apache.pig.Main - Apache Pig version 0.12.0-cdh5.13.0 (rexported) compiled O
ct 04 2017, 11:09:03
2026-03-09 23:15:56,231 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/cloudera/pig_177312335621
4.log
2026-03-09 23:15:57,190 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/cloudera/.pigbootup n
ot found
2026-03-09 23:15:57,334 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 23:15:57,334 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 23:15:57,334 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to h
adoop file system at: hdfs://quickstart.cloudera:8020
2026-03-09 23:15:57,642 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to m
ap-reduce job tracker at: localhost:8021
2026-03-09 23:15:57,674 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 23:15:57,674 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 23:15:57,705 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
```

```
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTIme      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime
AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1771910702594_0008  2       1       6       6       6       6       4       4       4       4       D,DD,X  HASH_
JOIN    hdfs://quickstart.cloudera:8020/tmp/temp1036947795/tmp127115342,

Input(s):
Successfully read 2 records from: "hdfs://quickstart.cloudera:8020/user/cloudera/csec/data2.txt"
Successfully read 2 records from: "hdfs://quickstart.cloudera:8020/user/cloudera/csec/data1.txt"

Output(s):
Successfully stored 2 records (30 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp1036947795/tmp127115342"

Counters:
Total records written : 2
Total bytes written : 30
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1771910702594_0008


2026-03-09 23:16:28,816 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher -
 Success!
2026-03-09 23:16:28,822 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 23:16:28,822 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
2026-03-09 23:16:28,822 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... wi
ll not generate code.
2026-03-09 23:16:28,833 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to pro
cess : 1
2026-03-09 23:16:28,833 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input path
s to process : 1
(1,2,3,1,3)
(4,5,6,4,6)
2026-03-09 23:16:28,909 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated
. Instead, use fs.defaultFS
2026-03-09 23:16:28,909 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depreca
ted. Instead, use mapreduce.jobtracker.address
[cloudera@quickstart ~]$
```

POST LAB:

1. Detail the procedures necessary for creating and implementing a Pig script.

2. What function do the LOAD and STORE operators serve in Pig?

3. In what manner does Pig carry out the execution of Pig Latin scripts on Hadoop internally?

4. What benefits does Pig offer for large-scale data processing when compared to directly writing MapReduce programs

(For Evaluator's use only)

Comment of the Evaluator(if Any)	Evaluator's Observation Marks Secured:_____out of _____ Signature of the Evaluator